PROJECT 17 · PHASE 1 · MACHINE LEARNING

Where will the fire be tomorrow?

Next-day wildfire spread prediction with engineered tabular features: a leakage-aware evaluation on NDWS

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THE TASK

For every 1 km pixel near an active fire, predict whether it burns tomorrow

18,545

patches of 64 × 64 km, US 2012–2020 (Kaggle NDWS, Huot et al. 2022)

12

input layers: terrain, weather, drought, vegetation, population, fire at day t

1.1%

of pixels are positive, so accuracy is useless: we report PR-AUC

Phase 1 = tabular ML: XGBoost / LightGBM on engineered features. Phase 2 = spatial deep learning on the same splits.

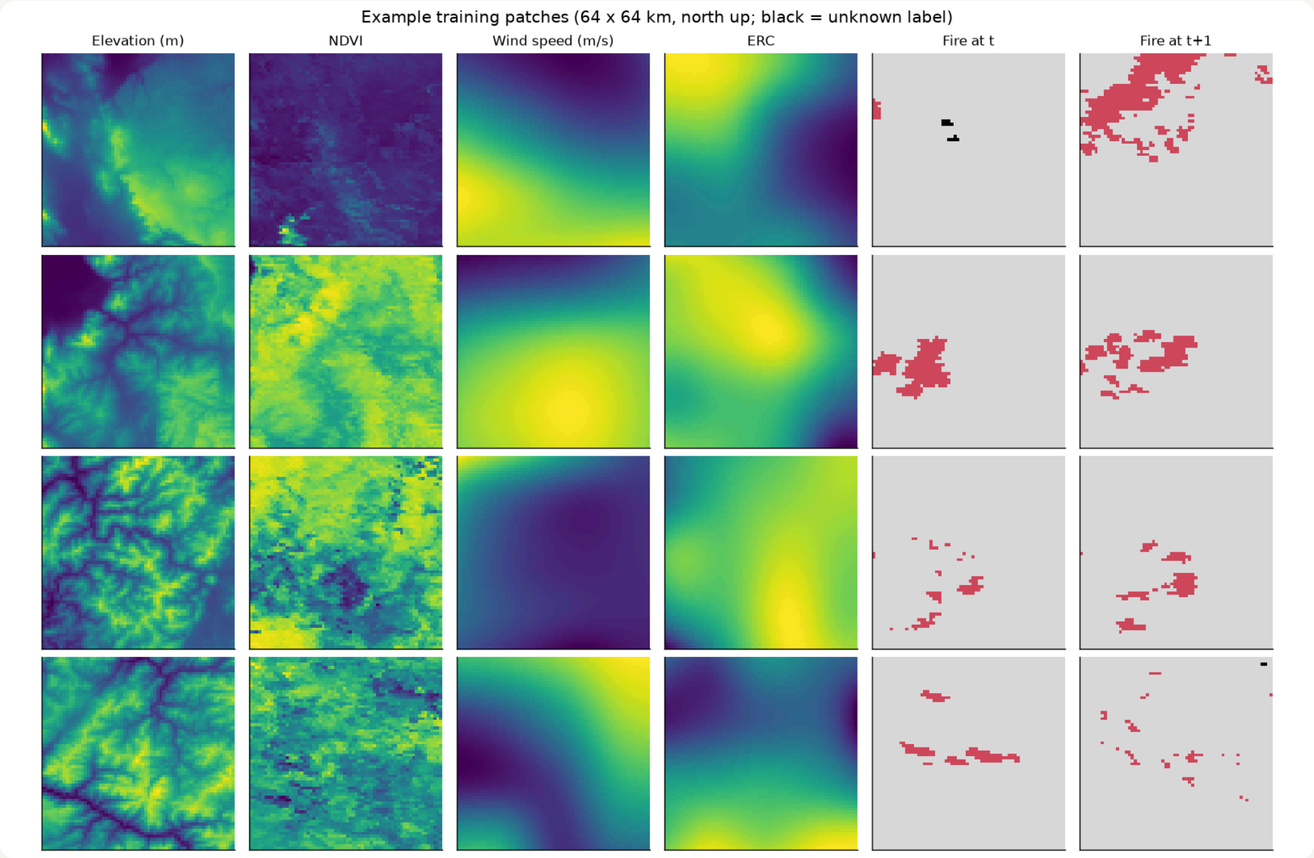
THE TEAM

Each member owns one module end to end

Member	Module	Code
Vedant Satbhai	Data access, spatial leakage audit	ndws.py, splits.py
Pratik Agade	EDA and data quality	eda.py, notebook 01
Lakshay Saharan	Feature engineering	features.py, tests
Raghvendra Singh	Models, imbalance, tuning	dataset.py, train.py
Aaryan Krishna	Evaluation, interpretation, report	metrics.py, evaluate.py

DATA

Each sample: inputs at day t, fire mask at day t+1



Official split: 14,979 / 1,877 / 1,689 patches, by random weeks

Labels: 1 fire · 0 no fire · −1 unknown (cloud), excluded

Cached as NumPy (4.3 GB) for fast, repeatable runs

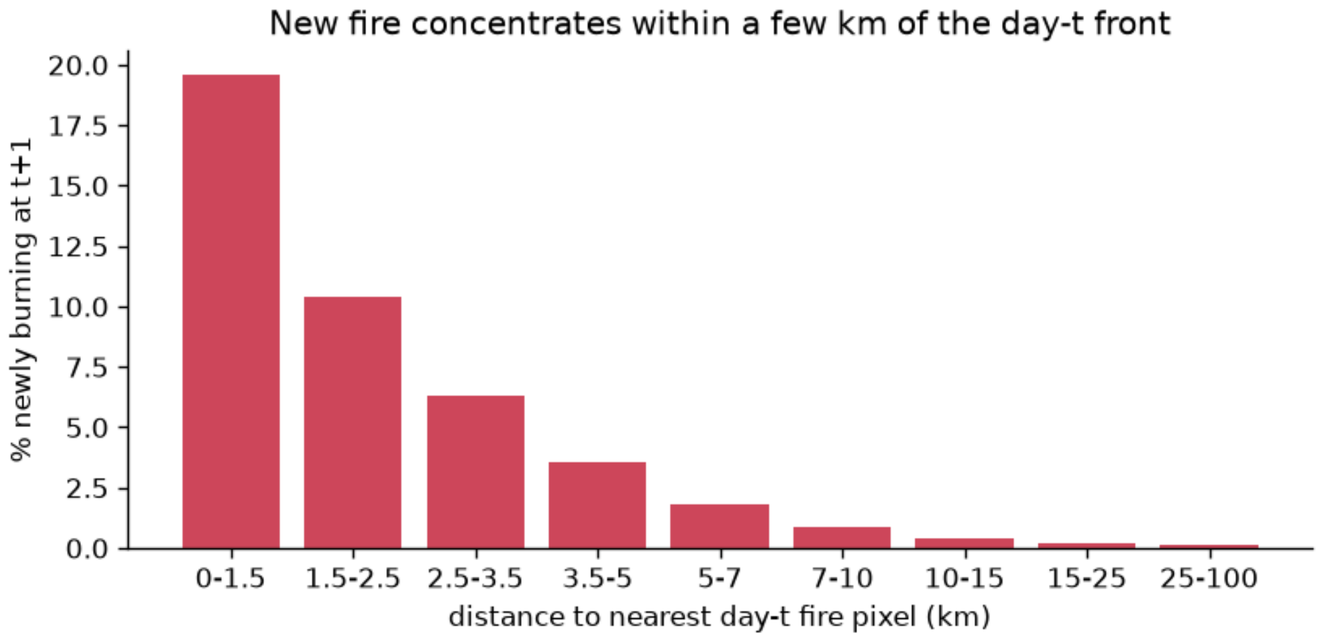
DATA QUALITY

The raw data contains physically impossible values

Found in the audit	Handling
Temperatures of 0 K and up to 344 K (71 °C)	Outside 200–330 K → missing
Wind directions from −496,522° to 37,620°	Outside 0–361° → missing
Negative humidity and wind speed	→ missing
Negative precipitation (interpolation)	Clipped at 0
Elevation slightly below sea level	Kept: real terrain
NDVI $\times 10^4$, heavy-tailed population	Rescaled, log1p

EDA

Most of tomorrow’s fire is new, and it starts near today’s front



74%

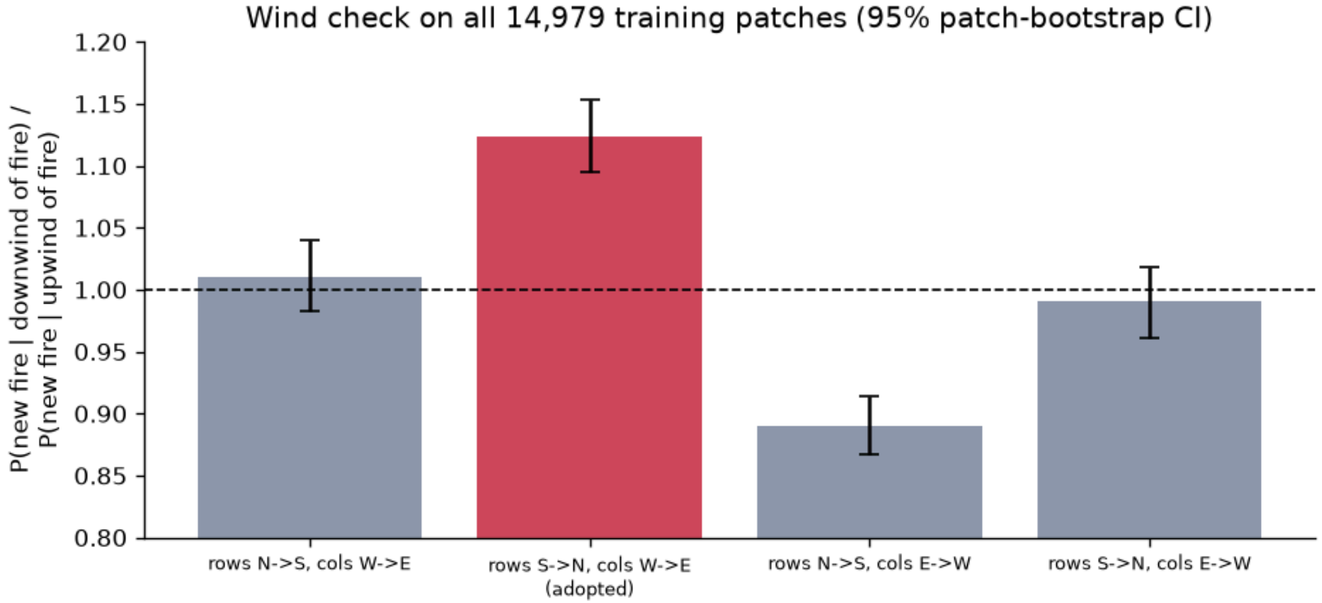
of day t+1 fire pixels were not burning at day t, so copying today’s mask fails

20%

ignition rate within 1.5 km of the front, vs 0.4% at 10–15 km

A PHYSICS TEST

The data has no map orientation, so we let the wind decide



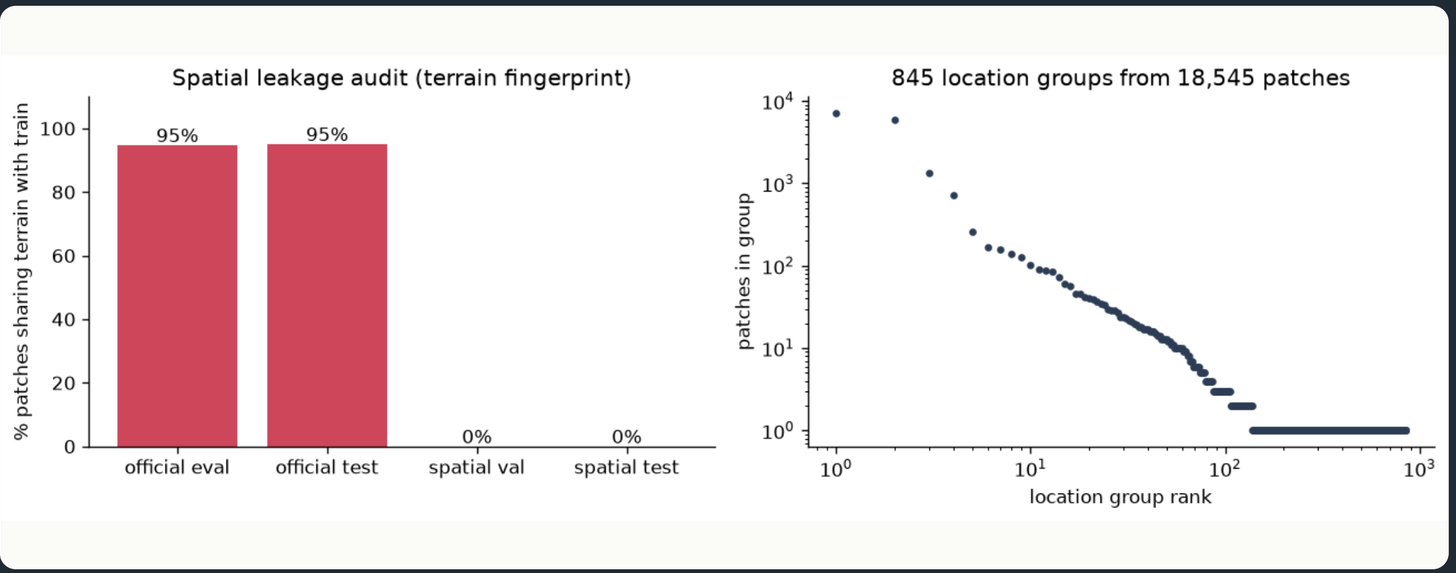
Fire should spread **downwind**. We tested all four row/column conventions on 14,979 patches.

Only rows south→north gives a real effect: **1.12×** (95% CI 1.09–1.15).

The usual assumption gives 1.01×: no wind effect at all.

SPATIAL LEAKAGE AUDIT

95% of official test patches cover the same ground as training



Fingerprint: elevation never changes, so an identical 8×8 km terrain block means the same place.

Union-find → 845 location groups. Control (mirrored): 0 / 1,855 matches.

New split: whole groups to train / val / test, 0% shared terrain.

FEATURE ENGINEERING

32 per-pixel features, all from day t only

Fire (8)

burning share in 3/5/9/17 km windows, distance to fire, fire count

Wind (6)

u, v; fire 1–3 km upwind or downwind; wind–fire alignment, wind push

Terrain (4)

slope, aspect, rise over run from the nearest fire pixel

Weather (2)

temperature range, vapour-pressure deficit (dryness)

Vegetation (2)

5 × 5 km NDVI mean and spread (fuel amount, continuity)

Raw (10)

cleaned elevation, weather, drought, NDVI, population, ERC

MODELS AND TRAINING

Four learners, two baselines, tuned only on validation

Models

Persistence and 3×3 persistence (no learning)
Logistic regression (L2)
Random forest
XGBoost, LightGBM with early stopping on val PR-AUC

Imbalance

All positives + 2% of negatives, train split only (1.9 M rows). Probabilities corrected back:
$p = r\cdot\hat{p} / (r\cdot\hat{p} + 1 - \hat{p}), r = 0.02$

A bug we fixed

First LightGBM stopped after 5 trees: pixels in a patch share weather, so small leaves memorise patches. Fix: leaves ≥ 2,000 rows, 50% column sampling.

RESULTS

We beat the published tabular baselines and approach the CNN

Test PR-AUC [95% CI]	Official split	Spatial split	Huot et al.
Persistence	0.107	0.037	0.115
Logistic regression	0.230	0.117	0.198
Random forest	0.265	0.135	0.225
XGBoost	**0.265** [0.248, 0.283]	**0.141** [0.128, 0.154]	–
LightGBM	0.263	0.138	–
CNN (Huot et al.)	–	–	0.284

Non-linearity matters.
The choice of tree ensemble does not.

+0.021

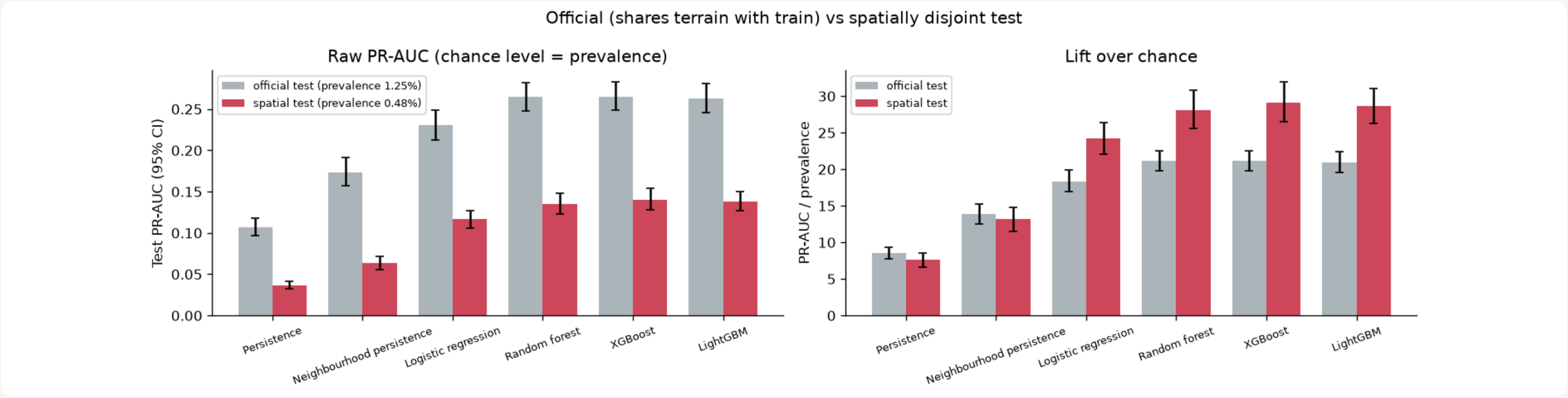
LightGBM over logistic regression (spatial), CI [+0.016, +0.026]: significant

+0.003

LightGBM over random forest, CI [−0.002, +0.006]: not significant

DOES THE OVERLAP INFLATE THE BENCHMARK?

Raw PR-AUC halves, but only because fires are rarer

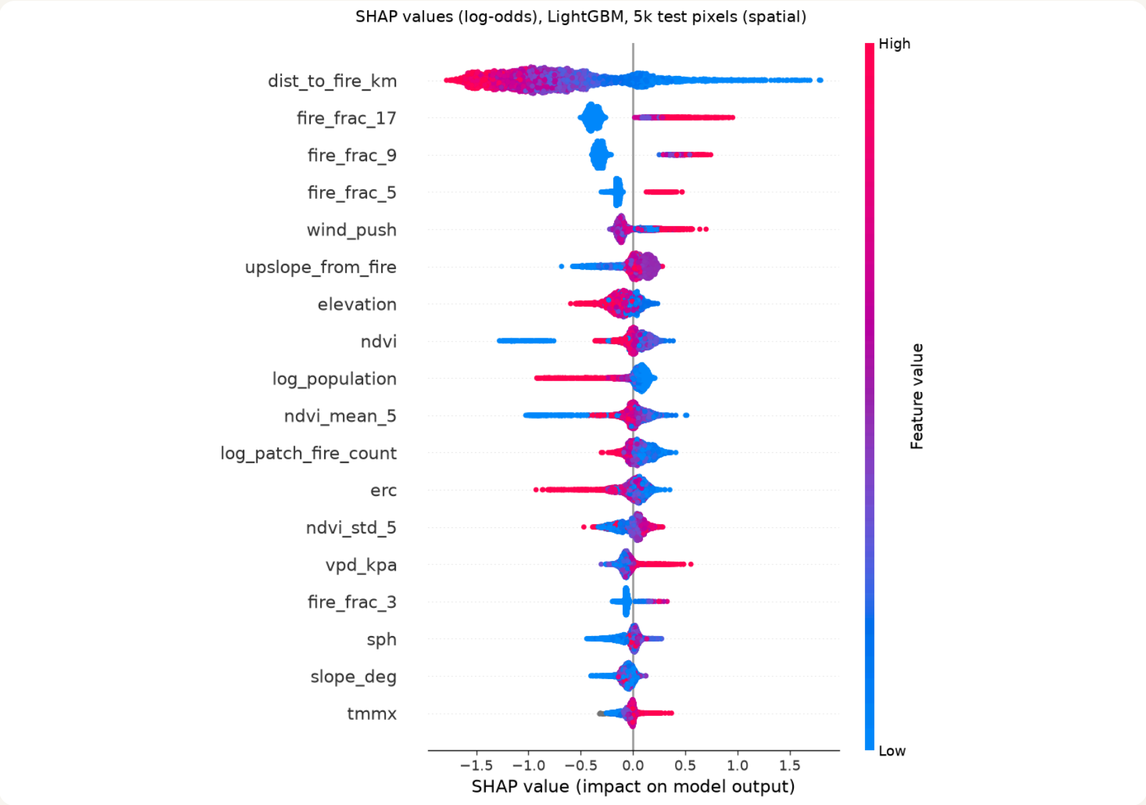


Spatial test prevalence 0.48% vs 1.25%. Lift over chance 28.6× vs 20.9×, ROC-AUC 0.896 vs 0.905, same model ranking: **no measurable inflation**.

WHAT DRIVES PREDICTIONS

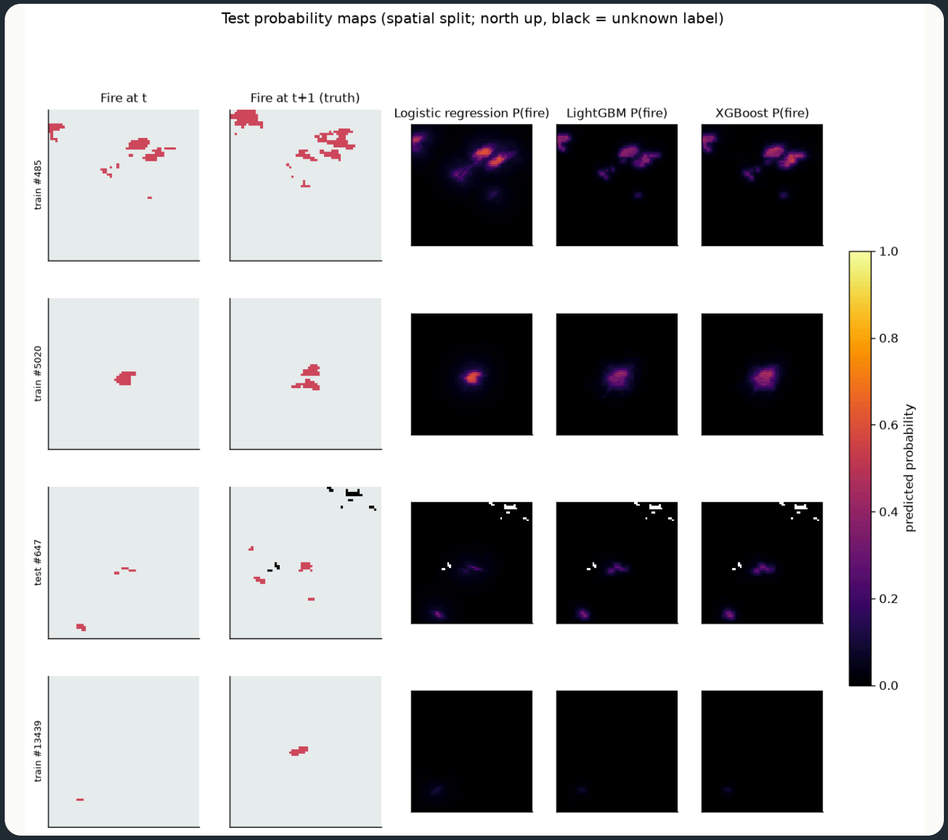
Where the fire is now matters most; wind comes next

LightGBM features	Test PR-AUC
Raw channels only	0.010
Fire features only	0.111
Raw + fire	0.130
+ wind	0.140
All 32	0.138



PROBABILITY MAPS

Tabular models draw halos, not fire fronts



Risk concentrates around the active front, and predicted rates match observed ones by distance (0.124 vs 0.117 within 1.5 km).

But blobs miss the front's shape and distant new ignitions (bottom row).

That gap is what Phase 2's U-Net must close.

LIMITATIONS AND NEXT STEPS

What limits us, and what Phase 2 must prove

Limitations

- Noisy daily MODIS labels: clouds, overpass timing
- No fuel moisture, fuel type, suppression or hourly wind
- Audit catches exact 8 × 8 km matches only
- Spatial test holds smaller fires than train

Phase 2 plan

- U-Net on the 12-channel patch, same splits and metrics
- ConvLSTM only with multi-day data (WildfireSpreadTS)
- Hybrid: engineered channels or LightGBM output into the U-Net
- Must beat XGBoost on the spatial test, paired test

SUMMARY

Engineered fire and wind features + tree ensembles: **0.265 PR-AUC**, above published tabular baselines.

95% of the official test shares terrain with train, yet our results hold on a location-disjoint split.

Questions?